A1

* $Y_S \odot \log p$

means element by element multiplication of matrices & addition of all such products

We have,

$$L = -\frac{1}{N} \underbrace{Y_S \odot \log p}_{l_1} + \underbrace{\frac{\lambda}{2}\left(\|W_1\|^2 + \dots\right)}_{l_2}$$

For $\frac{\partial L}{\partial S}$, since t, W_1, W_2, W_3 are constant

We have, $\frac{\partial L}{\partial S} = \frac{\partial l_1}{\partial S}$

~~We want~~

Also, $P = \text{softmax}(S)$ (rowwise)

~~Using softmax formula,~~

S is of shape $(N \times C)$

So, shape of P is $(N \times C)$

Since softmax is now used

We have $P_{ij} = \frac{e^{S_{ij}}}{\sum_{k=1}^{C} e^{S_{ik}}}$

eqn too

When doing log of matrix, each element gets logged

$$\log(P_{ij}) = S_{ij} - \log\left(\sum_{k=1}^{C} e^{S_{ik}}\right)$$

$$Y_S = (1-\varepsilon)Y + \frac{\varepsilon}{C}$$

$\therefore Y_S$ is shape of Y which is also CNxC

eqn $\therefore l_i = -\frac{1}{N} \sum_{i,j=(1,1)}^{i,j=(N,C)} \left[(1-\varepsilon)Y_{ij} + \frac{\varepsilon}{C}\right]\left[S_{ij} - \log\left(\sum_{k=1}^{C} e^{S_{ik}}\right)\right]$

$$l_i = -\frac{1}{N} \sum \Big[(1-\varepsilon)Y_{ij}S_{ij} + \frac{\varepsilon}{C} S_{ij} - (1-\varepsilon)Y_{ij} \log\left(\sum_{k=1}^{C} e^{S_{ik}}\right) - \frac{\varepsilon}{C} \log\left(\sum_{k=1}^{C} e^{S_{ik}}\right) \Big]$$

$$l_i = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{s_{ij}} \log p_{ij}$$

$$l_i = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{s_{ij}} \left[s_{ij} - \log \left(\sum_{k=1}^{C} e^{s_{ik}} \right) \right]$$

$$l_i = -\frac{1}{N} \left[\sum_{i=1}^{N} \left(\sum_{j=1}^{C} y_{s_{ij}} s_{ij} \right) - \sum_{j=1}^{C} y_{s_{ij}} \log \sum_{k=1}^{C} e^{s_{ik}} \right]$$

Depends only on row ∴ ~~can cons~~ Is constant for each column

$$l = -\frac{1}{N} \left[\sum_{i=1}^{N} \sum_{j=1}^{C} y_{s_{ij}} s_{ij} - \sum_{i=1}^{N} \log \sum_{k=1}^{C} e^{s_{ik}} \sum_{j=1}^{C} y_{s_{ij}} \right]$$

$$\therefore l_f = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{s_{ij}} s_{ij}$$

$$+ \frac{1}{N} \sum_{i=1}^{N} \left[\left(\log \sum_{k=1}^{C} e^{s_{ik}} \right) \cdot \sum_{j=1}^{C} y_{s_{ij}} \right]$$

It is given y_s rows sum to one

$$\therefore \sum_{j=1}^{C} y_{s_{ij}} = 1$$

$$\rightarrow L_i = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{s_{ij}} s_j + \frac{1}{N} \sum_{i=1}^{N} \log \left(\sum_{\pi=1}^{C} e^{s_{i\pi}} \right)$$

↓ P

↓ 2

This is basically

$$\log(e^{s_{11}} + e^{s_{12}} + e^{s_{13}})$$

$$+ \log(e^{s_{21}} + e^{s_{22}} + e^{s_{23}})$$

⋮

We will diff this element wise.

Say differentiating for s_{ab}, other s_{ij} are constant.

so this diff becomes 0

Only $y_{s_{ab}} s_{ab}$ is left in summation

whose derivative is $y_{s_{ab}}$

$$\therefore \frac{\partial p}{\partial s_{ab}} = -\frac{1}{N} y_{s_{ab}}$$

$\frac{\partial q}{\partial S_{ab}}$ = for this, all logs that having a^{th} row and constant only $\log\left(\sum_{k=1} e^{(S_{ak})}\right)$ survives

$$\therefore \frac{\partial q}{\partial S_{ab}} = \frac{1}{N} \frac{e^{S_{ab}}}{\log \sum_{k=1} e^{S_{ak}}} = \frac{1}{N} P_{ab}$$

$$\frac{\partial L}{\partial S_{ab}} = \frac{\partial p}{\partial S_{ab}} + \frac{\partial q}{\partial S_{ab}}$$

$$= -\frac{1}{N} Y_{S_{ab}} + \frac{1}{N} P_{ab}$$

$$= \frac{1}{N} (P_{ab} - Y_{S_{ab}})$$

Now, since $\frac{\partial L}{\partial S}$ is a matrix of shape of S whose $(a,b)^{th}$ element is

$$\frac{1}{N} (P_{ab} - Y_{S_{ab}})$$

And is true for every element, we have

$$\frac{\partial L}{\partial S} = \frac{1}{N}\left(P_{ab} - Y_{S_{ab}}\right)$$

Hence Proved

$$\frac{\partial L}{\partial V_{ij}} = \frac{\partial L}{\partial H_{m_{ij}}} \times \frac{\partial H_m}{\partial V_{ij}}$$

upstream × local

$$H_m = V \odot G$$

$$\frac{\partial H_{m_{ij}}}{\partial [\,]_{ij}} = G_{ij} \Rightarrow \frac{\partial L}{\partial V_{ij}} = \frac{\partial L}{\partial H_{m_{ij}}} \times G_{ij}$$

For all elements:-

$$\frac{\partial L}{\partial V} = \frac{\partial L}{\partial H_m} \odot G = dH_m \odot G$$

$$G = \sigma(V), \quad \sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)\left[1 - \sigma(x)\right]$$

$$\frac{\partial L}{\partial V} = \frac{\partial L}{\partial H_m} \odot \sigma'(V)$$

$$\frac{\partial L}{\partial V_{ij}} = \frac{\partial L}{\partial H_{m_{ij}}} \times \frac{\partial H_{m_{ij}}}{\partial V_{ij}}$$

$$H_{m_{ij}} = V_{ij} \odot \sigma(V_{ij}), \quad \frac{\partial H_{m_{ij}}}{\partial V_{ij}} = V_{ij} \times \sigma'(V_{ij})$$

$$= V_{ij} \times G_{ij}\left[1 - G_{ij}\right]$$

$$\frac{\partial L}{\partial V_{ij}} = dH_{m_{ij}} * V_{ij} * G_{ij} * (1 - G_{ij})$$

$$\frac{\partial L}{\partial V} = dH_m \odot V \odot G \odot (1 - G)$$

A2 $n_i = \sqrt{\|H_{m_i}\|^2 + 0.0001}$ (approx)

↳ basically $\|H_i\|_2$

Basically n_i is the square root of sum of squares of each element in i^{th} row of H_m matrix

$$R_{ia} = H_{m_{ia}} / n_i$$

Basically, R_i is matrix of shape $(1, H)$

i.e matrix with each element of ith row divided by n_i

$\frac{\partial R_i}{\partial H_{m_j}}$ will give us jacobian matrix of size (H, H)

R_i

Say,

Hm is (N, H)

Say $H = 3$

$\therefore R_i = \left[Hm_{i_1} \rightarrow Hm_{i_2}, Hm_{i_3} \right.$

$\sqrt{Hm_{i_1}^2 + Hm_{i_2}^2 + H}$

$$R_i = \left[\frac{Hm_{i_1}}{n_i}, \frac{Hm_{i_2}}{n_i}, \frac{Hm_{i_3}}{n_i} \right].$$

where n_i is $= \sqrt{Hm_{i_1}^2 + Hm_{i_2}^2 + Hm_{i_3}^2}$ (approx)

$\therefore$ Each element of R_i depends on all $Hm_{i_1}, Hm_{i_2}, Hm_{i_3}$...

$\therefore$ We need, $\frac{\partial R_{i_a}}{\partial Hm_{i_b}}$ for all $a, b \in \{1, 2, 3\}$

$\hookrightarrow \therefore$ A jacobian matrix we will have as $\frac{\partial R}{\partial H}$ of size (3,3)

$$\frac{\partial R_i}{\partial H_{m_i}} = \begin{bmatrix} \frac{\partial R_{i_1}}{\partial Hm_{i_1}} & \frac{\partial R_{i_2}}{\partial Hm_{i_2}} & \frac{\partial R_{i_3}}{\partial Hm_{i_3}} \\ \frac{\partial R_{i_2}}{\partial Hm_{i_1}} & \sim & \sim \\ \sim & \sim & \sim \end{bmatrix}$$

Basically have 2nd elem of R depends on 2nd elem of H

Q2

Q1b+c+d+2

We have,

$$R_{ij} = \frac{H_{ij}}{N_i^2 \left[H_{ij} + H_{ik} + \dots \right]}$$

Case-1 $j \neq k$

$$\frac{\partial R_{ij}}{\partial H_{mn}} = \frac{H_{ij}}{h_i^2}$$

Case-2 $j = k$

$$\frac{\partial R_{ij}}{\partial H_{mn}} = \frac{1}{h_i^2} - \frac{H_{ij}}{h_i^2}$$

R_{ij} ... H_m

$dR = \frac{\partial R}{\partial H_m} \cdot \frac{dH_m}{dx}$

Case-1 $j \neq k$

$\frac{dR_{ij}}{dH_{mik}}$ is like $\frac{d\left(\frac{d}{\sqrt{c+x^2}}\right)}{dx}$ where $d = H_{ij}$

$c = n_i^2 - H_{mik}^2$

$x = H_{ik}$

$= -d(c+x^2)^{-\frac{1}{2}}$

$= -\frac{x \cdot d}{(c+x^2)^{3/2}}$

$\therefore \frac{dR_{ij}}{dH_{mik}} = \frac{H_{ik} \cdot H_{ij}}{n_i \cdot n_i \cdot n_i} = -\frac{R_{ik} \cdot R_{ij}}{n_i}$

Case-2 $j = k$

$\frac{dR_{ij}}{dH_{mij}}$ is like $\frac{d\left(\frac{x}{\sqrt{c+x^2}}\right)}{dx} = \frac{c}{(c+x^2)^{3/2}}$

$\therefore \frac{dR_{ij}}{dH_{ij}} = \frac{n_i^2 - H_{mij}^2}{n_i^3} \Rightarrow \frac{1}{n_i}(1 - R_{ij}^2)$

Let our

Our Jacobian matrix is all $\frac{\partial R_{ik}}{\partial H_{mij}}$ element

Let $\delta_{kj} = \begin{cases} 1 & \text{if } k=j \\ 0 & \text{if } k \neq j \end{cases}$

$\therefore J_{kj}^{(i)} = \frac{\partial R}{\partial h_k} \times \frac{\partial R_{ik}}{\partial H_{mij}} = \frac{1}{h_i}(\delta_{kj} - R_{ik} R_{ij})$

We are supposed to find $\frac{\partial L}{\partial H_{mij}}$

L depends on R

& each element of R_i depends on all elements of H_{mi}

$\therefore$ R_i changes if any element of H_{mi} changes

$\therefore$ L changes if any elmnt of H_{mi} chngs

Because of so many contributing changes

We say, $\frac{\partial L}{\partial H_{ij}} = \sum_{k=1}^{H} \frac{\partial L}{\partial R_{ik}} \left(\frac{\partial R_{ik}}{\partial H_{ij}}\right)$

$\downarrow$

J_{kj}

$$\frac{\partial L}{\partial H_{ij}} = \sum_{k=1}^{H} \frac{\partial R_{ik}}{\partial} \cdot \frac{1}{n_i} \left(\delta_{kj} - R_{ik} \cdot R_{ij} \right)$$

some fast thing

some fast thing

$$= \frac{1}{n_i} \left(dR_i \right)$$

$$= \frac{1}{n_i} \left[\sum_{k=1}^{H} \left(\frac{\partial}{\partial} R_{ik} \delta_{kj} - \frac{\partial}{\partial} R_{ik} \cdot R_{ij} \right) \right]$$

$$\delta_{kj} = 0 \quad \text{if} \quad k \neq j$$

$$\sum \partial R_{ik} \cdot \delta_{kj} = \partial R_{ij} \text{ only}$$

$$= \frac{1}{n_i} \left[\frac{\partial}{\partial} R_{ij} - R_{ij} \sum R_{ik} \frac{\partial}{\partial} R_{ik} \right]$$

dot product

$$= \frac{1}{n_i} \left[dR_{ij} - R_{ij} \odot dR_i \right]$$

Hadamard product

$$\downarrow$$

$$dHim_{ij}$$

* If $1e^{-6}$ part is removed, $\|R_i\| = 1$ but we don't use this in our derivation

Q4 Since, X is used in two branches, so their gradients are added

$$\frac{\partial L}{\partial X} = dUW_1^T + dVW_2^T$$

Computational graph :-

